

Lab Sheet 6 – MPI
IT24101500

EX1 – Hello World

EX2 -

np=1 total=50000005000000 time=0.042230

np=1 total=50000005000000 time=0.038097

np=1 total=50000005000000 time=0.039165

np=1 total=50000005000000 time=0.038825

np=1 total=50000005000000 time=0.037745

np=1 total=50000005000000 time=0.038870

np=1 total=50000005000000 time=0.044470

np=1 total=50000005000000 time=0.038358

np=1 total=50000005000000 time=0.039660

np=1 total=50000005000000 time=0.040386

np=2 total=50000005000000 time=0.021166

np=2 total=50000005000000 time=0.021821

np=2 total=50000005000000 time=0.020730

np=2 total=50000005000000 time=0.017362

np=2 total=50000005000000 time=0.020585

np=2 total=50000005000000 time=0.022383

np=2 total=50000005000000 time=0.021795

np=2 total=50000005000000 time=0.022557

np=2 total=50000005000000 time=0.022839

np=2 total=50000005000000 time=0.022195

np=4 total=50000005000000 time=0.013193

np=4 total=50000005000000 time=0.011366

np=4 total=50000005000000 time=0.013264

np=4 total=50000005000000 time=0.011249

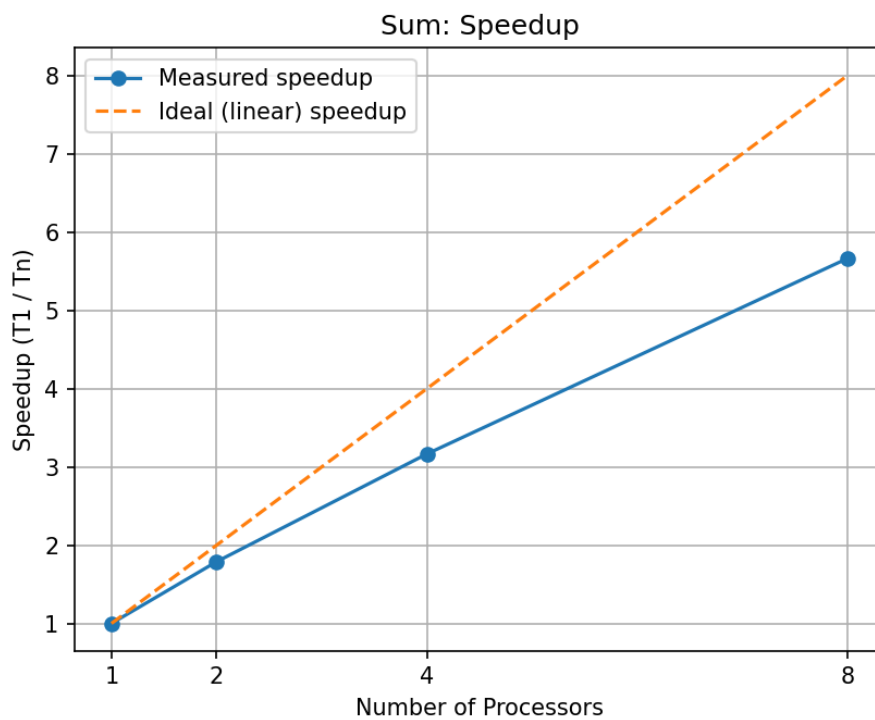
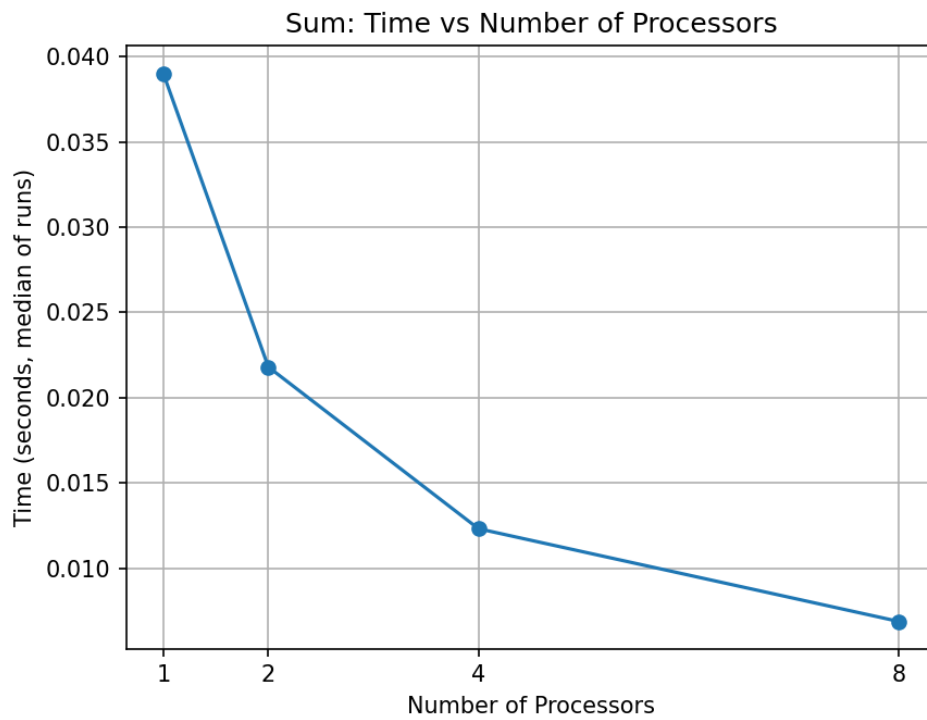
np=4 total=50000005000000 time=0.013225
np=4 total=50000005000000 time=0.011459
np=4 total=50000005000000 time=0.013250
np=4 total=50000005000000 time=0.008817
np=4 total=50000005000000 time=0.013515
np=4 total=50000005000000 time=0.011159
np=8 total=50000005000000 time=0.010748
np=8 total=50000005000000 time=0.006868
np=8 total=50000005000000 time=0.008056
np=8 total=50000005000000 time=0.007611
np=8 total=50000005000000 time=0.006828
np=8 total=50000005000000 time=0.006828
np=8 total=50000005000000 time=0.006817
np=8 total=50000005000000 time=0.006911
np=8 total=50000005000000 time=0.008079
np=8 total=50000005000000 time=0.006728

Ex3 -

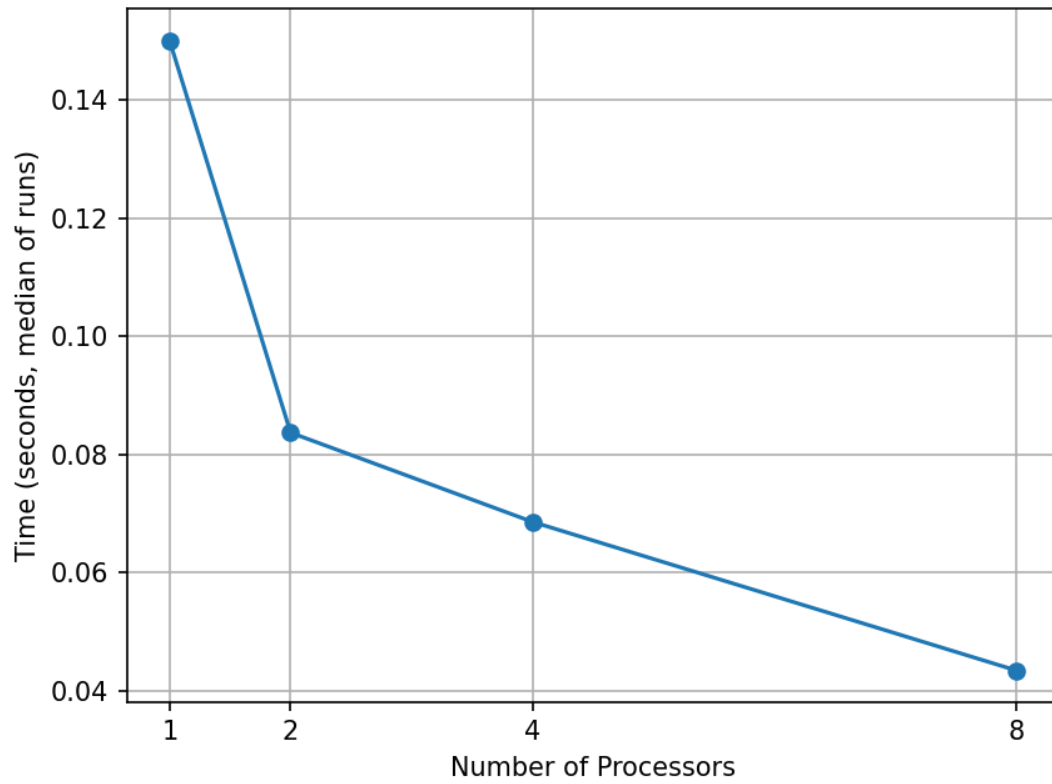
np=1 pi=3.141386 time=0.151259
np=1 pi=3.141386 time=0.150364
np=1 pi=3.141386 time=0.150554
np=1 pi=3.141386 time=0.149097
np=1 pi=3.141386 time=0.148332
np=1 pi=3.141386 time=0.148101
np=1 pi=3.141386 time=0.149660
np=2 pi=3.141265 time=0.095807
np=2 pi=3.141265 time=0.087026
np=2 pi=3.141265 time=0.084474

np=2 pi=3.141265 time=0.082847
np=2 pi=3.141265 time=0.082219
np=2 pi=3.141265 time=0.082972
np=2 pi=3.141265 time=0.081460
np=2 pi=3.141265 time=0.100435
np=2 pi=3.141265 time=0.081472
np=2 pi=3.141265 time=0.085935
np=4 pi=3.140876 time=0.059839
np=4 pi=3.140876 time=0.076221
np=4 pi=3.140876 time=0.061102
np=4 pi=3.140876 time=0.077502
np=4 pi=3.140876 time=0.073547
np=4 pi=3.140876 time=0.069602
np=4 pi=3.140876 time=0.061027
np=4 pi=3.140876 time=0.069130
np=4 pi=3.140876 time=0.065074
np=4 pi=3.140876 time=0.068113
np=8 pi=3.141124 time=0.062932
np=8 pi=3.141124 time=0.043421
np=8 pi=3.141124 time=0.043349
np=8 pi=3.141124 time=0.044146
np=8 pi=3.141124 time=0.041059
np=8 pi=3.141124 time=0.042738
np=8 pi=3.141124 time=0.063815
np=8 pi=3.141124 time=0.043437
np=8 pi=3.141124 time=0.043541
np=8 pi=3.141124 time=0.043514

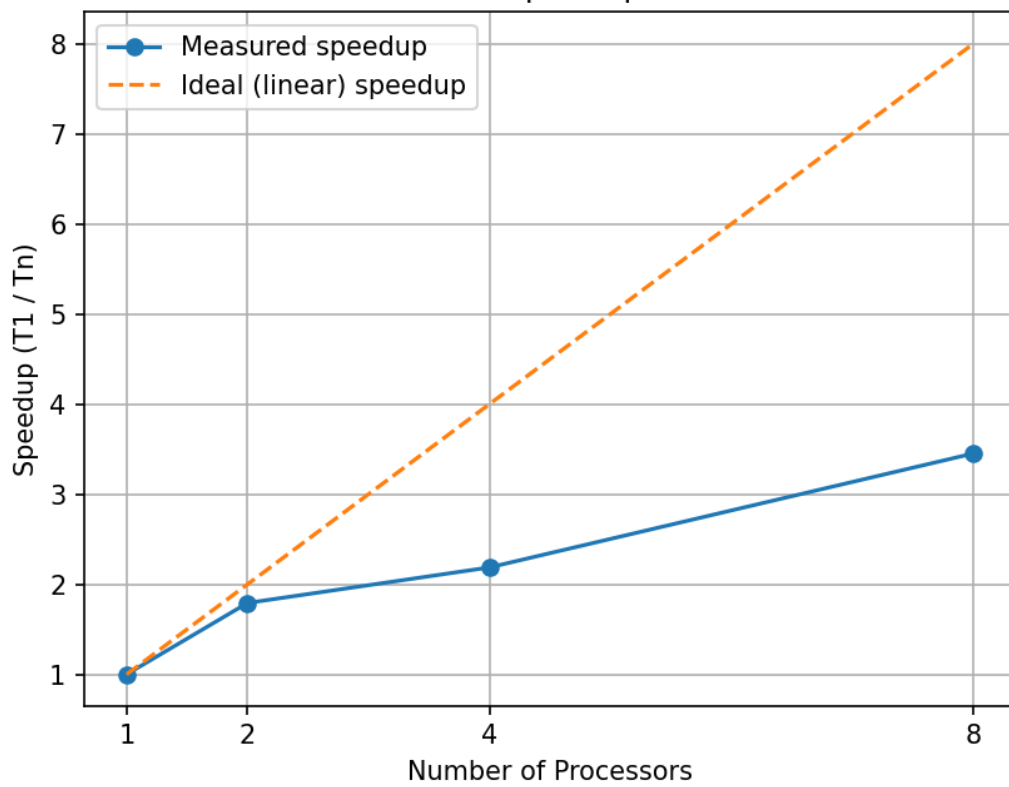
Ex 4 -



Pi: Time vs Number of Processors



Pi: Speedup



Ex5 -

```
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08/ex5$ mpicc mismatch.c -o mismatch
timeout 5 mpirun --oversubscribe -np 4 ./mismatch
Rank 1: sending x to rank 3
Rank 3: waiting for a message from rank 2
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08/ex5$
```

Rank 1 sends to rank 3, but rank 3 is waiting for a message from rank 2. Rank 2 never sends anything, so rank 3's MPI_Recv blocks forever.

Rank 1 used MPI_Ssend (synchronous send), which only completes once the matching receive has started. Rank 3's receive never matches, so rank 1 blocks as well. That's why "Rank 1: send finished" never printed.

```
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08/ex5$ mpicc bsend.c -o bsend
mpirun --oversubscribe -np 4 ./bsend
Rank 0: just a normal process
Rank 1: buffered-sending x to rank 3
Rank 1: overwrote x with -1 after Bsend
Rank 2: just a normal process
Rank 3: received y = 0 10 20 30 40 50 60 70 80 90
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08/ex5$
```

MPI_Bsend copies the data into the attached buffer before returning, so the sender can reuse or overwrite x immediately without corrupting the message in flight.

EX 6 -

```
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08$ cd ex6 && ./run_pi_any.sh && cd ..
python3 compare_ex6.py
np=1 pi=3.141386 time=0.151742 order=
np=1 pi=3.141386 time=0.150237 order=
np=1 pi=3.141386 time=0.149375 order=
np=1 pi=3.141386 time=0.150369 order=
np=1 pi=3.141386 time=0.150979 order=
np=1 pi=3.141386 time=0.149574 order=
np=1 pi=3.141386 time=0.148632 order=
np=1 pi=3.141386 time=0.150552 order=
np=1 pi=3.141386 time=0.150917 order=
np=1 pi=3.141386 time=0.149941 order=
np=2 pi=3.141265 time=0.081308 order=1
np=2 pi=3.141265 time=0.081652 order=1
np=2 pi=3.141265 time=0.081858 order=1
np=2 pi=3.141265 time=0.081525 order=1
np=2 pi=3.141265 time=0.078320 order=1
np=2 pi=3.141265 time=0.082968 order=1
np=2 pi=3.141265 time=0.080618 order=1
np=2 pi=3.141265 time=0.102870 order=1
np=2 pi=3.141265 time=0.081146 order=1
np=2 pi=3.141265 time=0.082292 order=1
np=4 pi=3.140876 time=0.060648 order=2 1 3
np=4 pi=3.140876 time=0.060146 order=2 1 3
np=4 pi=3.140876 time=0.081640 order=3 1 2
np=4 pi=3.140876 time=0.076426 order=3 1 2
np=4 pi=3.140876 time=0.072547 order=3 2 1
np=4 pi=3.140876 time=0.071470 order=2 3 1
np=4 pi=3.140876 time=0.069022 order=3 1 2
np=4 pi=3.140876 time=0.069336 order=2 3 1
np=4 pi=3.140876 time=0.072349 order=1 2 3
np=4 pi=3.140876 time=0.064966 order=1 2 3
np=8 pi=3.141124 time=0.042689 order=4 6 5 1 7 2 3
np=8 pi=3.141124 time=0.042380 order=5 3 6 7 1 4 2
np=8 pi=3.141124 time=0.033373 order=2 1 4 5 6 3 7
np=8 pi=3.141124 time=0.040571 order=6 1 2 3 4 5 7
np=8 pi=3.141124 time=0.042574 order=6 1 2 3 5 7 4
np=8 pi=3.141124 time=0.042375 order=5 1 6 3 2 4 7
np=8 pi=3.141124 time=0.040819 order=7 6 3 5 2 1 4
np=8 pi=3.141124 time=0.043218 order=2 3 4 5 6 7 1
np=8 pi=3.141124 time=0.041132 order=7 1 3 6 5 4 2
np=8 pi=3.141124 time=0.052257 order=7 4 1 5 2 6 3

np | pi (Ex 3) | pi (Ex 6) | median time (Ex 3) | median time (Ex 6)
1 | 3.141386 | 3.141386 | 0.150073 | 0.150303
2 | 3.141265 | 3.141265 | 0.083723 | 0.081589
4 | 3.140876 | 3.140876 | 0.068622 | 0.070403
8 | 3.141124 | 3.141124 | 0.043476 | 0.042377
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08$
```

Arrival Order -

np=8 pi=3.141124 time=0.042027 order=3 2 4 6 1 7 5

np=8 pi=3.141124 time=0.038843 order=3 1 2 5 6 7 4

np=8 pi=3.141124 time=0.041792 order=4 6 1 7 3 2 5

np=8 pi=3.141124 time=0.041916 order=1 2 4 5 6 7 3

np=8 pi=3.141124 time=0.040232 order=5 1 6 2 7 4 3

Ex-7

```

vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08$ cd ex7 && ./run_pi_bsend.sh && cd ..
python3 compare_ex7.py
np=1 pi=3.141386 time=0.154910 order=
np=1 pi=3.141386 time=0.150147 order=
np=1 pi=3.141386 time=0.148143 order=
np=1 pi=3.141386 time=0.148485 order=
np=1 pi=3.141386 time=0.151201 order=
np=1 pi=3.141386 time=0.150627 order=
np=1 pi=3.141386 time=0.150090 order=
np=1 pi=3.141386 time=0.151073 order=
np=1 pi=3.141386 time=0.164928 order=
np=1 pi=3.141386 time=0.151330 order=
np=2 pi=3.141265 time=0.082705 order=1
np=2 pi=3.141265 time=0.081000 order=1
np=2 pi=3.141265 time=0.080708 order=1
np=2 pi=3.141265 time=0.082997 order=1
np=2 pi=3.141265 time=0.081555 order=1
np=2 pi=3.141265 time=0.082056 order=1
np=2 pi=3.141265 time=0.082262 order=1
np=2 pi=3.141265 time=0.081566 order=1
np=2 pi=3.141265 time=0.081604 order=1
np=2 pi=3.141265 time=0.080228 order=1
np=4 pi=3.140876 time=0.060496 order=1 2 3
np=4 pi=3.140876 time=0.070003 order=3 1 2
np=4 pi=3.140876 time=0.074485 order=2 3 1
np=4 pi=3.140876 time=0.066942 order=1 2 3
np=4 pi=3.140876 time=0.061458 order=1 3 2
np=4 pi=3.140876 time=0.069785 order=1 2 3
np=4 pi=3.140876 time=0.074759 order=3 1 2
np=4 pi=3.140876 time=0.060401 order=1 3 2
np=4 pi=3.140876 time=0.059984 order=2 1 3
np=4 pi=3.140876 time=0.068449 order=1 3 2
np=8 pi=3.141124 time=0.042656 order=5 1 4 6 7 2 3
np=8 pi=3.141124 time=0.041639 order=7 4 5 6 3 2 1
np=8 pi=3.141124 time=0.057532 order=2 7 5 3 6 1 4
np=8 pi=3.141124 time=0.040661 order=6 1 2 3 4 5 7
np=8 pi=3.141124 time=0.042416 order=6 1 2 3 4 5 7
np=8 pi=3.141124 time=0.041584 order=5 1 6 3 4 2 7
np=8 pi=3.141124 time=0.041376 order=5 7 3 1 6 4 2
np=8 pi=3.141124 time=0.042105 order=1 7 5 3 2 6 4
np=8 pi=3.141124 time=0.042435 order=1 2 3 4 5 6 7
np=8 pi=3.141124 time=0.035299 order=1 2 4 5 7 6 3
- - - - -
np | pi (Ex 6) | pi (Ex 7) | median time (Ex 6) | median time (Ex 7)
1 | 3.141386 | 3.141386 | 0.150303 | 0.150850
2 | 3.141265 | 3.141265 | 0.081589 | 0.081585
4 | 3.140876 | 3.140876 | 0.070403 | 0.067695
8 | 3.141124 | 3.141124 | 0.042377 | 0.041872
vimukthi@VK-Linux:~/Downloads/PC-Lab-08/IT24101500-Herath-PC-08$

```